

Standard quantum mechanics

Cauchy (initial-value) problem

$\psi(x, t_0)$ given $\rightarrow$ evolve forward

$$i\hbar \partial_t \psi = H\psi$$

Schrodinger / Dirac equation

$$|\psi|^2 = \text{Born rule}$$

Fundamental axiom

Time: fundamental coordinate

Lorentzian spacetime a priori

Measurement / update rule

Textbook formulations; interpretations differ

Euclidean Condensate Theory

Specified BVP / ensemble

Existence / uniqueness / selection: Open

$$\delta^{\wedge}(AB) \partial_A \partial_B \Phi - V'(\Phi) = 0$$

Euclidean condensate equation

Admitted-measure representation

dim ≥ 3 : $C_1 + C_2$ projectors;
dim 2: $C_1 + C_2 q$ effects
state / event / probability /
outcome / update: Open

P4 supplies an ordered direction

Scalar signature needs supplied coefficients;
selection / physical cones: Open

Measurement / outcome map: Open

Decoherence alone does not select outcome